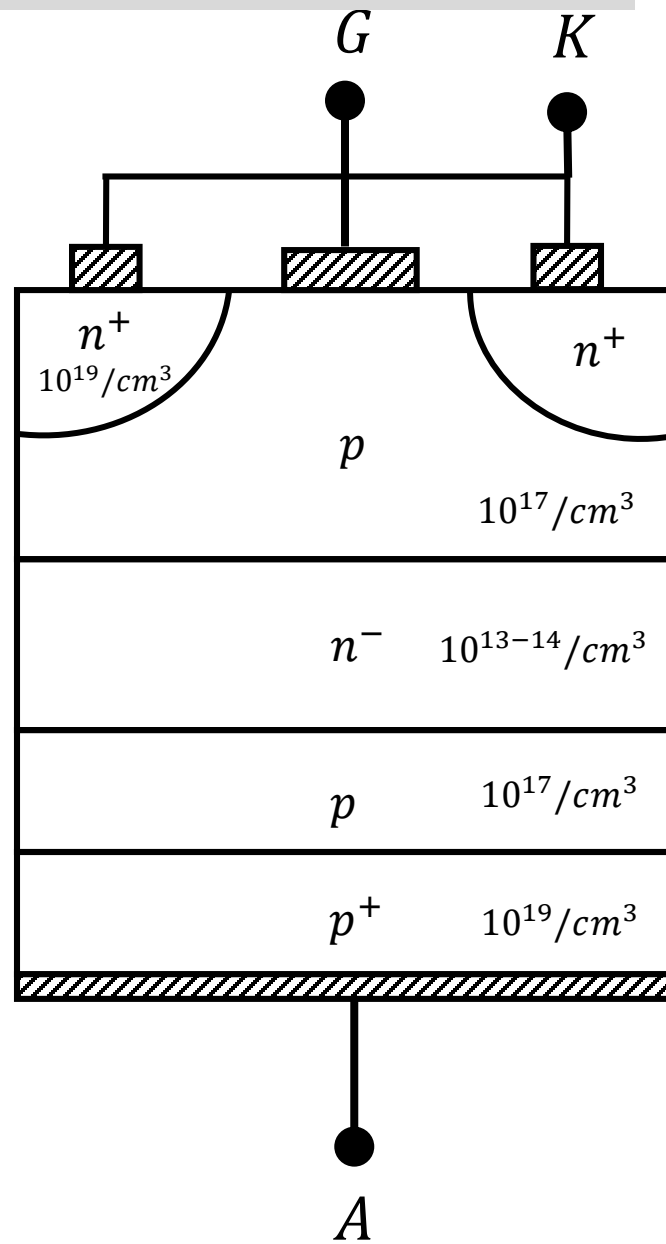
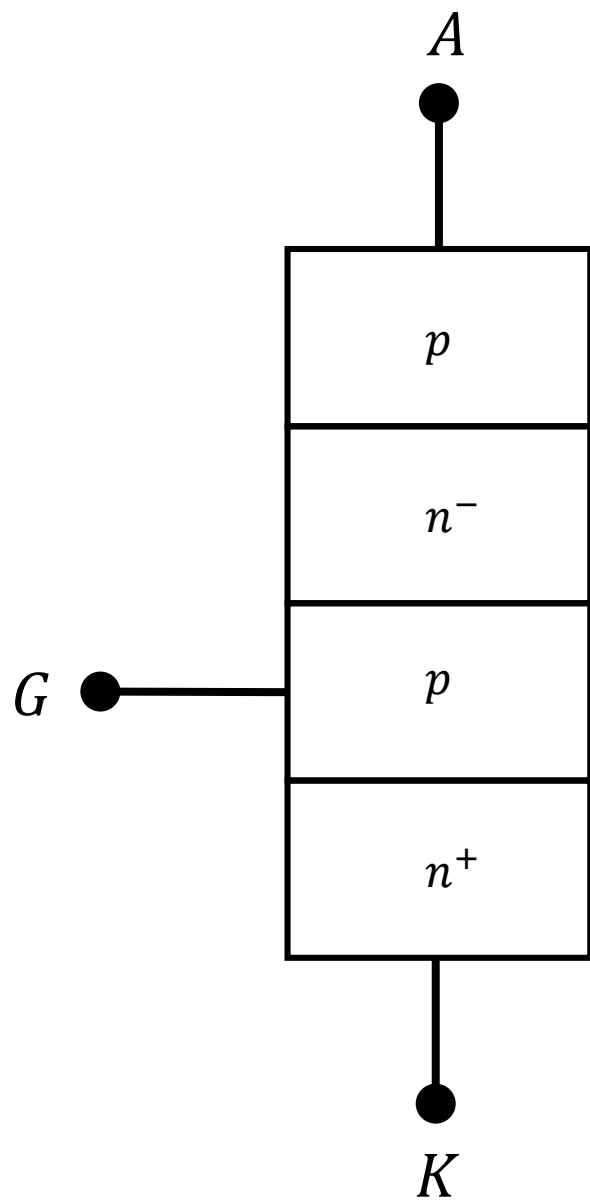
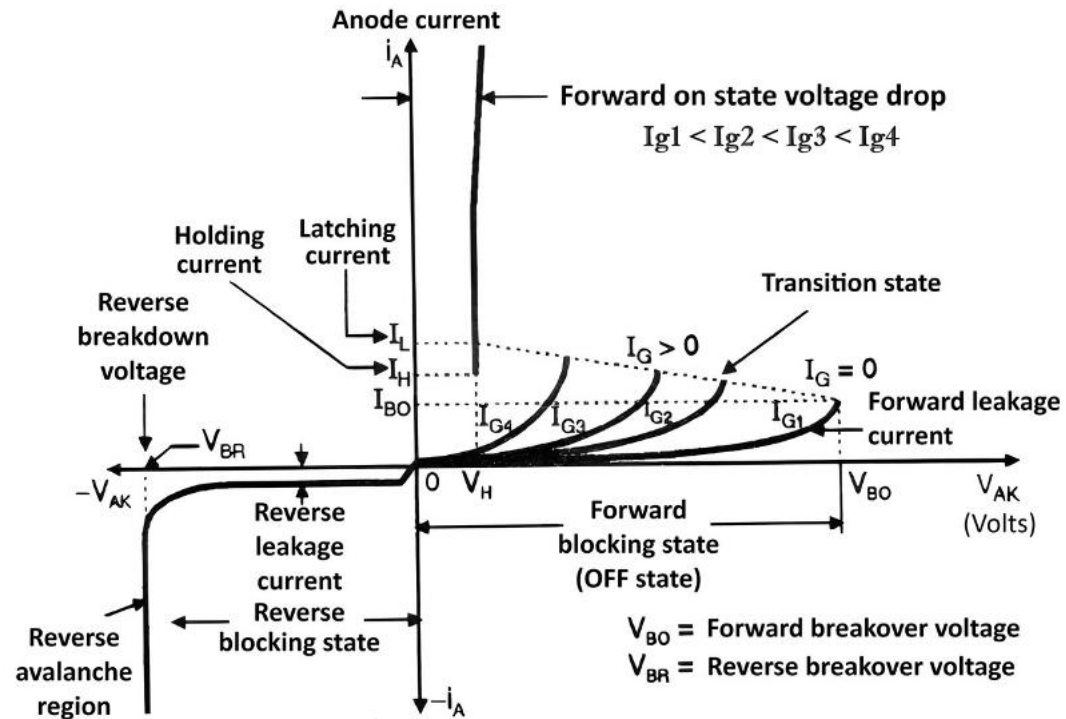


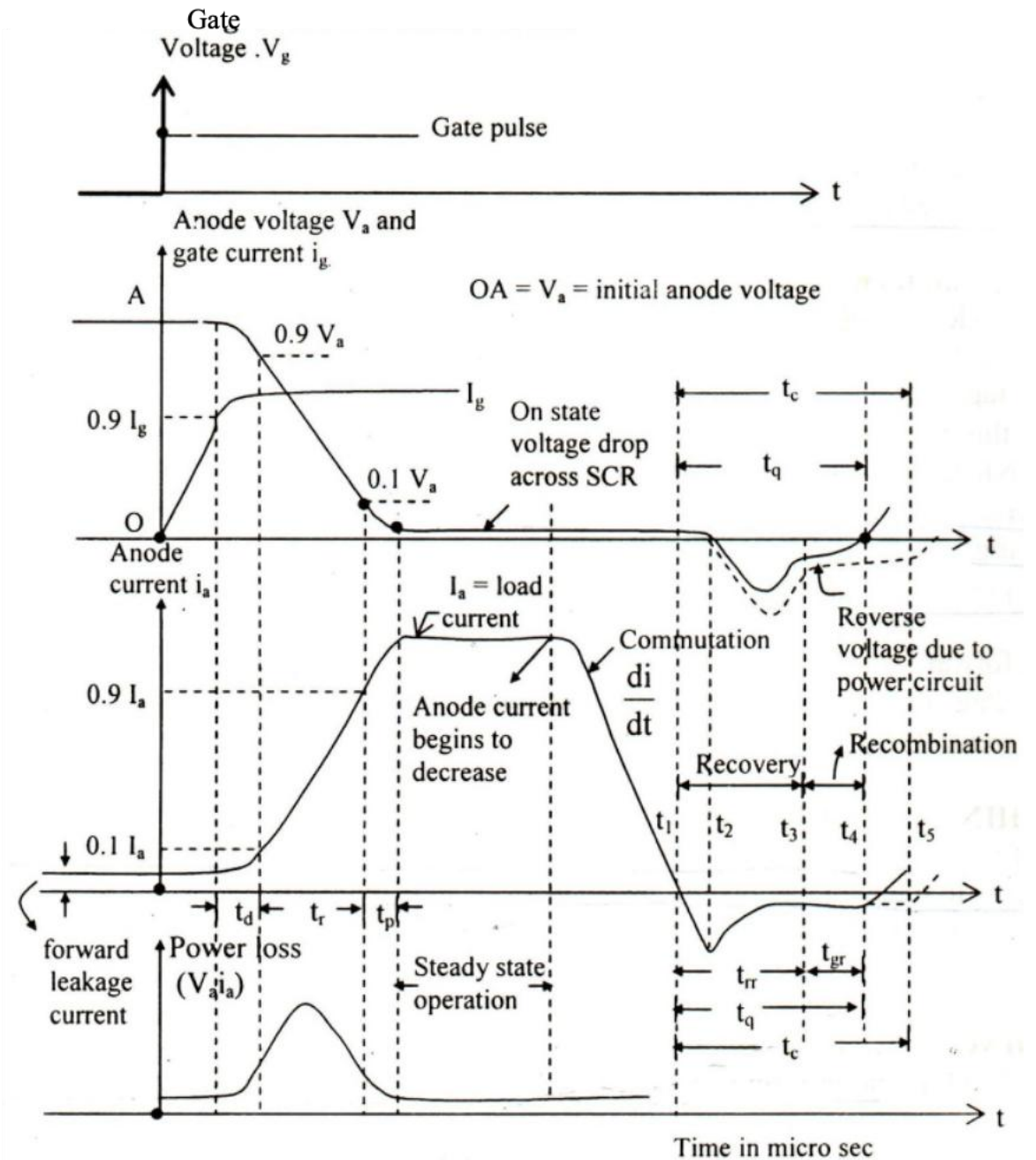
SCR (silicon controlled rectifier)



IV and switching characteristics of an SCR or Thyristor

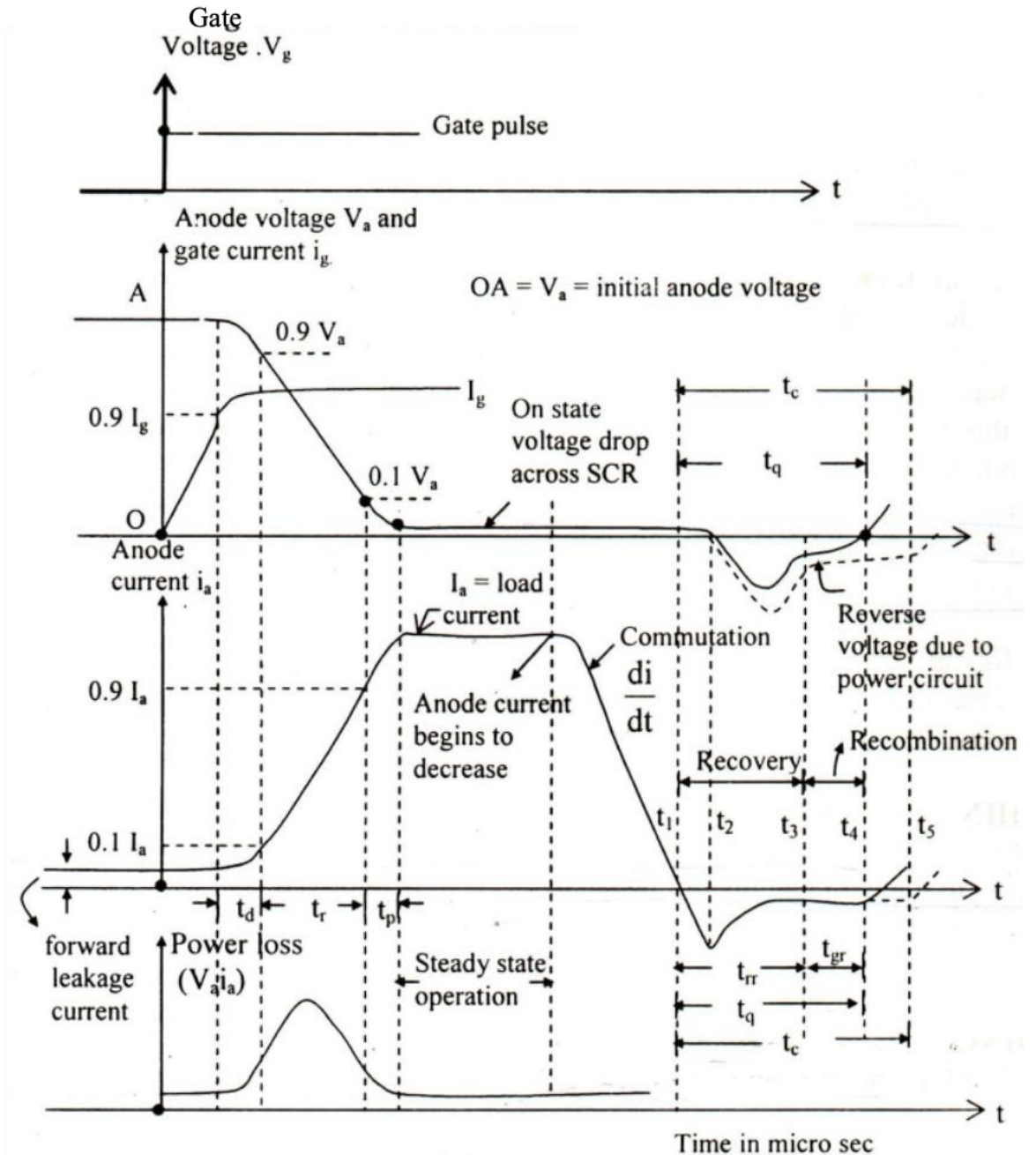


V-I Characteristics of SCR



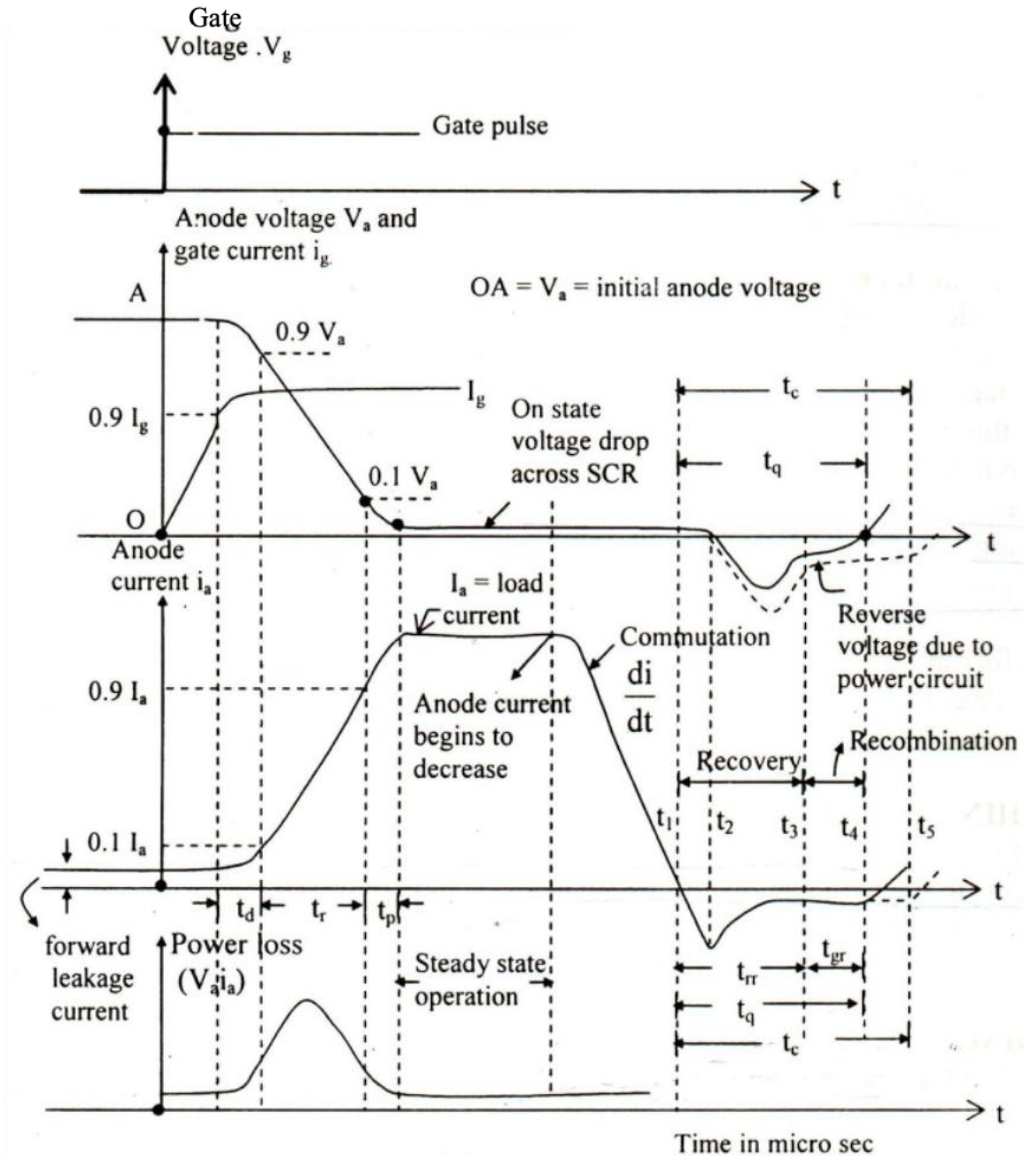
Turn-On Time

- ❖ A forward-biased thyristor is usually turned on by applying a positive gate voltage between gate and cathode.
- ❖ There is, however, a transition time from forward off-state to forward on state.
- ❖ This transition time, called thyristor turn-on time, is defined as the time during which it changes from forward blocking state to final on-state.
- ❖ Total turn-on time can be divided into three intervals;
 - delay time t_d ,
 - rise time t_r and
 - spread time t_p



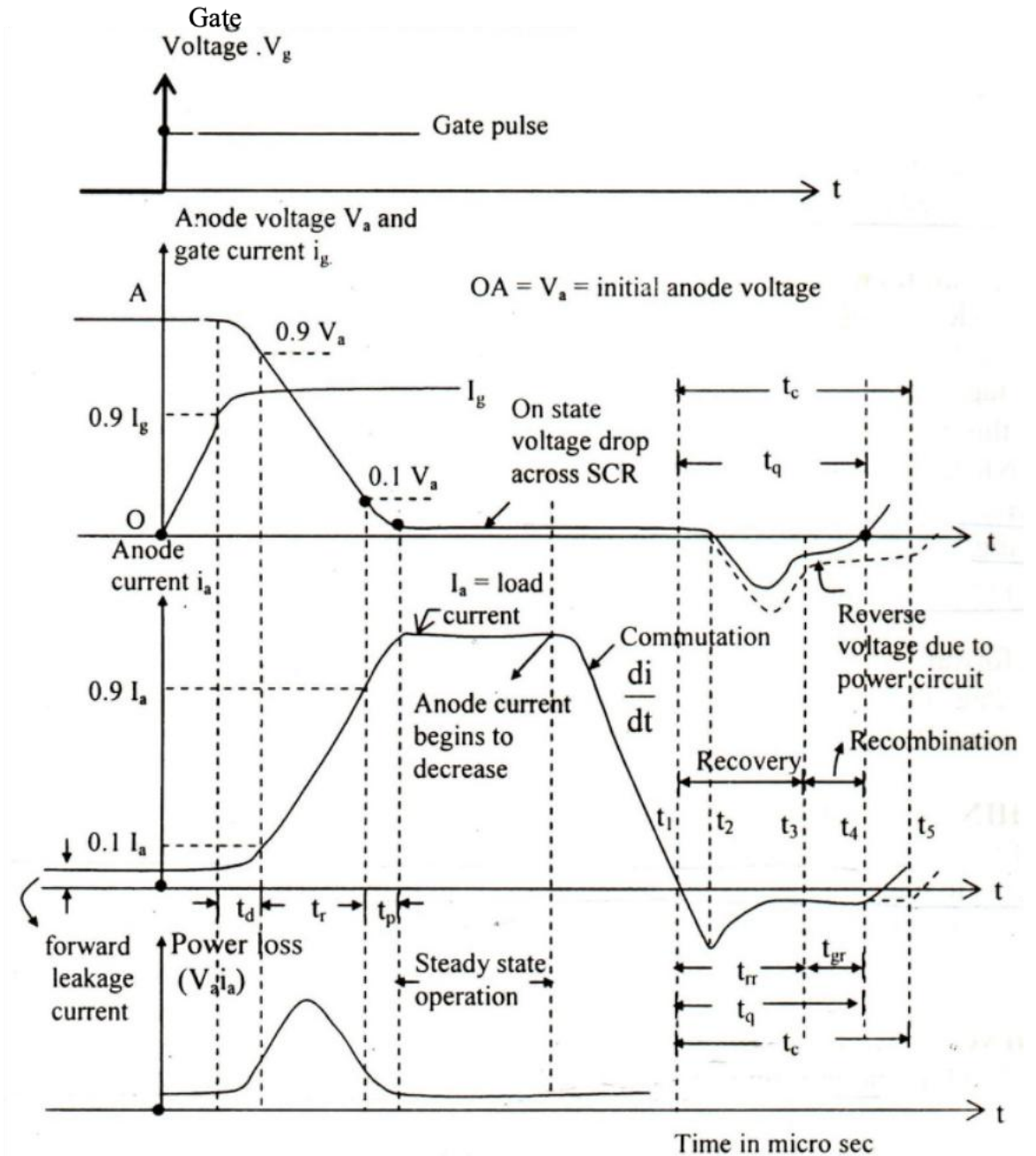
Delay Time t_d

- ❖ The delay time t_d is measured from the instant at which gate current reaches $0.9I_g$ to the instant at which anode current reaches $0.1I_a$. I_g and I_a are the final values of gate and anode currents respectively.
- ❖ The delay time may also be defined as the time during which anode voltage falls from V_a to $0.9V_a$ where V_a = initial value of anode voltage.
- ❖ Another way of defining delay time is the time during which anode current rises from forward leakage current to $0.1I_a$ where I_a = final value of anode current.
- ❖ The delay time can be decreased by applying high gate current and more forward voltage between anode and cathode.
- ❖ The delay time is fraction of a microsecond.



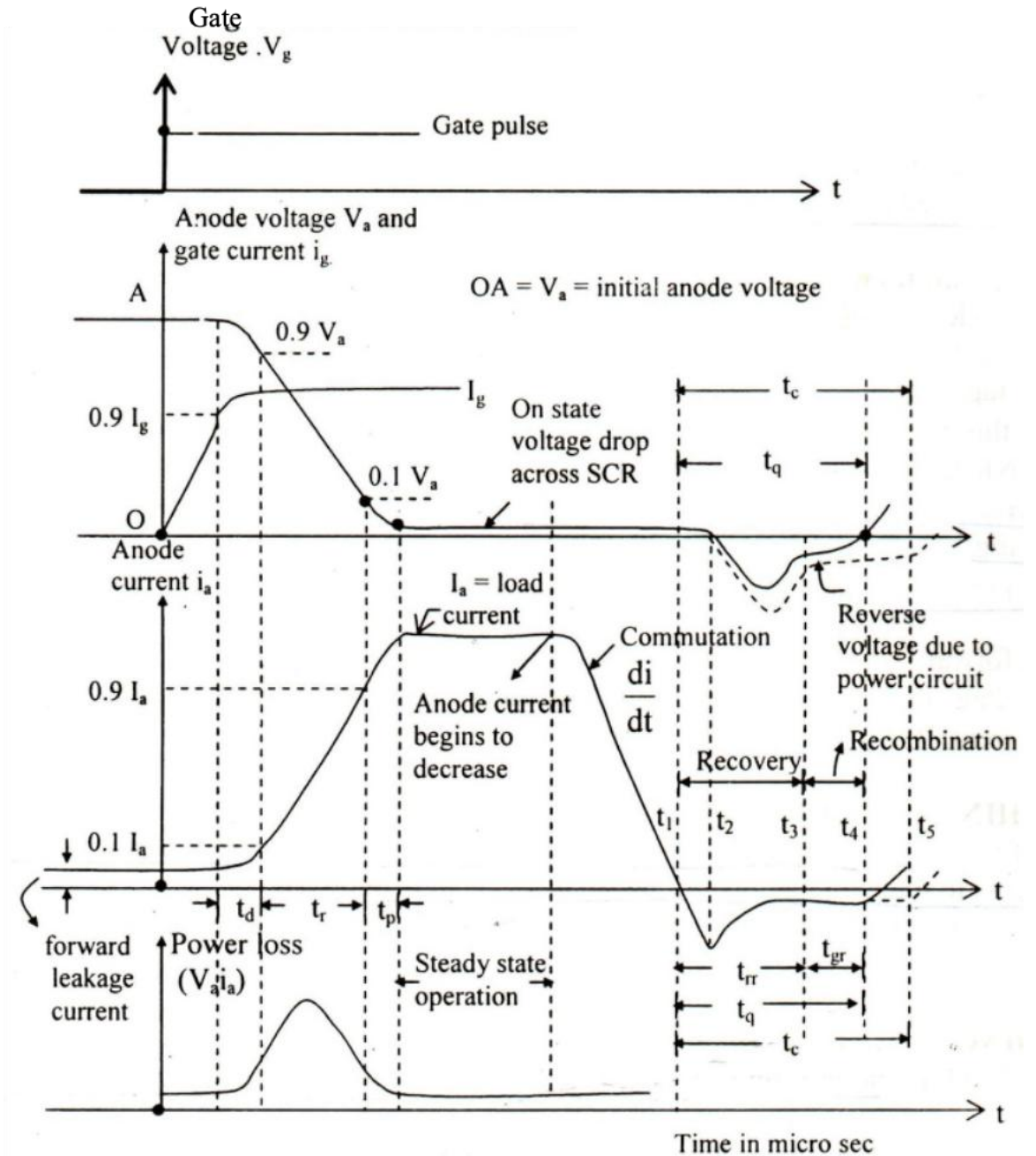
Rise Time t_r

- ❖ The rise time t_r is the time taken by the anode current to rise from $0.1I_a$ to $0.9I_a$; I_a is the final values of anode current.
- ❖ The rise time may also be defined as the time during which anode voltage falls from $0.9V_a$ to $0.1V_a$ where V_a = initial value of anode voltage.
- ❖ The rise time is inversely proportional to the magnitude of gate current and its build up rate.
- ❖ Thus, t_r can be reduced if high and steep current pulses are applied to the gate. However, the main factor determining t_r is the nature of anode circuit.
- ❖ For example, for series RL circuit, the rate of rise of anode current is slow, therefore, t_r is more. For RC series circuit, $\frac{di}{dt}$ is high, t_r is therefore, less.



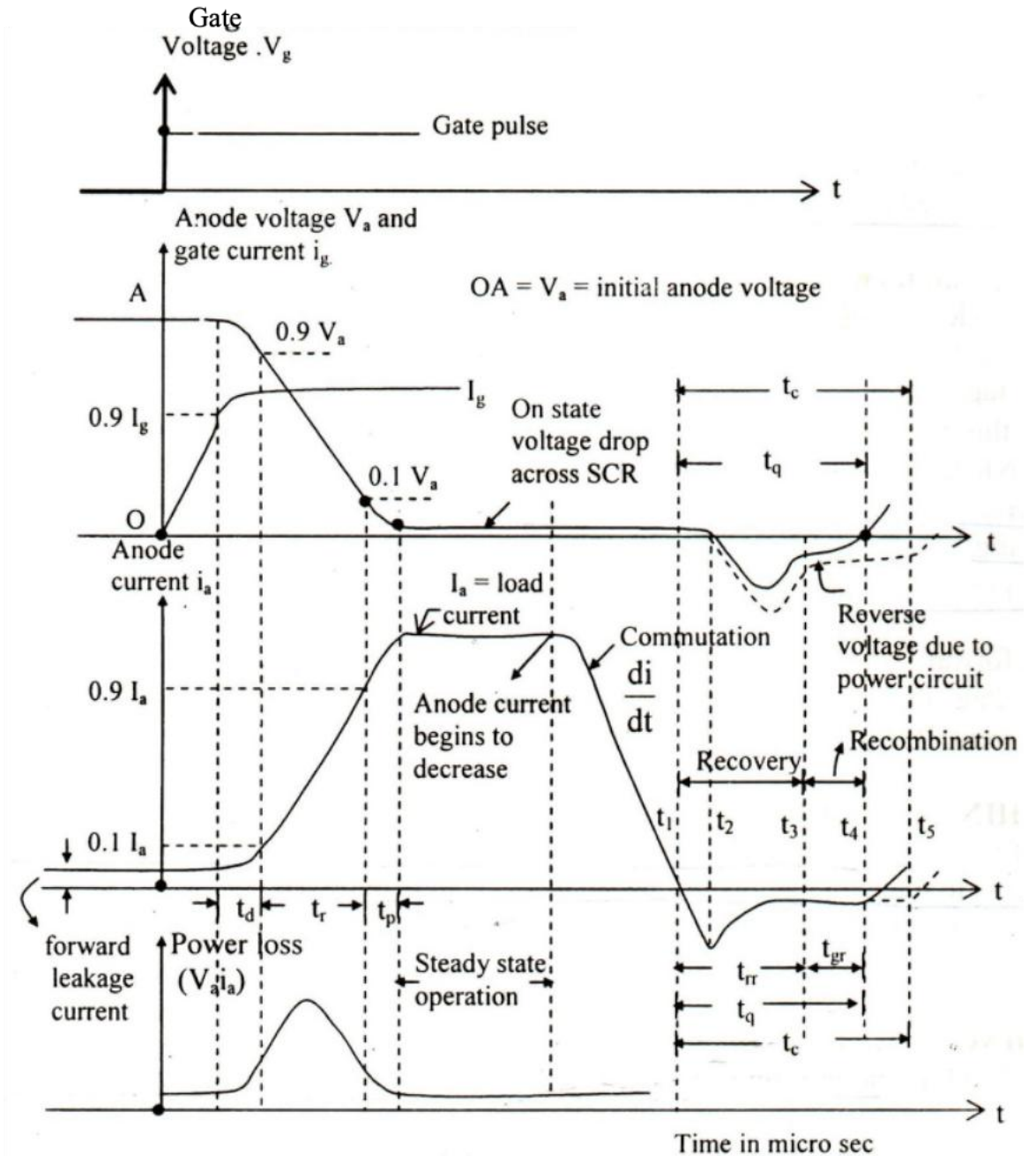
Spread Time t_p

- ❖ The spread time t_p is the time taken by the anode current to rise from $0.9I_a$ to I_a ; I_a is the final values of anode current.
- ❖ The spread time may also be defined as the time during which anode voltage falls from $0.1V_a$ to *on-state voltage drop* ; where V_a = initial value of anode voltage.
- ❖ During this time, conduction spreads over the entire cross-section of the cathode of SCR.
- ❖ The spreading interval depends on the area of cathode and on gate structure of the SCR.
- ❖ After the spread time, anode current attains steady state value, and the voltage drop across SCR is equal to the on-state voltage drop of the order of 1 to 1.5 V.



Turn off Time

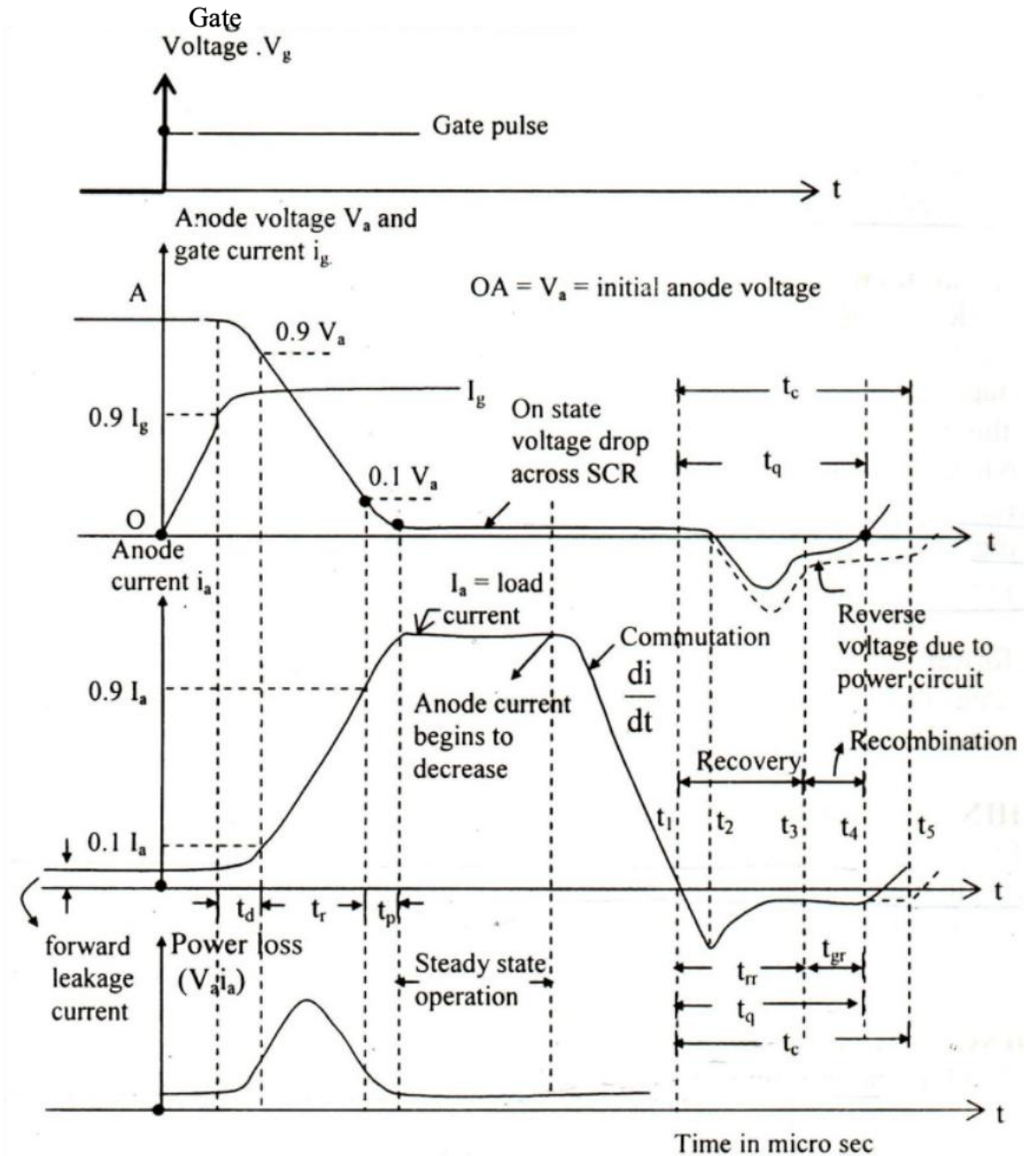
- ❖ Thyristor turn-off means that it has changed from on to off state and is capable of blocking the forward voltage.
- ❖ This dynamic process of the SCR from conduction state to forward blocking state is called commutation process or turn-off process.
- ❖ Once the thyristor is on, gate loses control. The SCR can be turned off by reducing the anode current below holding current.
- ❖ If forward voltage is applied to the SCR at the moment its anode current falls to zero, the device will not be able to block this forward voltage as the carriers (holes and electrons) in the four layers are still favourable for conduction. The device will therefore go into conduction immediately even though gate signal is not applied.
- ❖ In order to obviate such an occurrence, it is essential that the thyristor is reverse biased for a finite period after the anode current has reached zero.



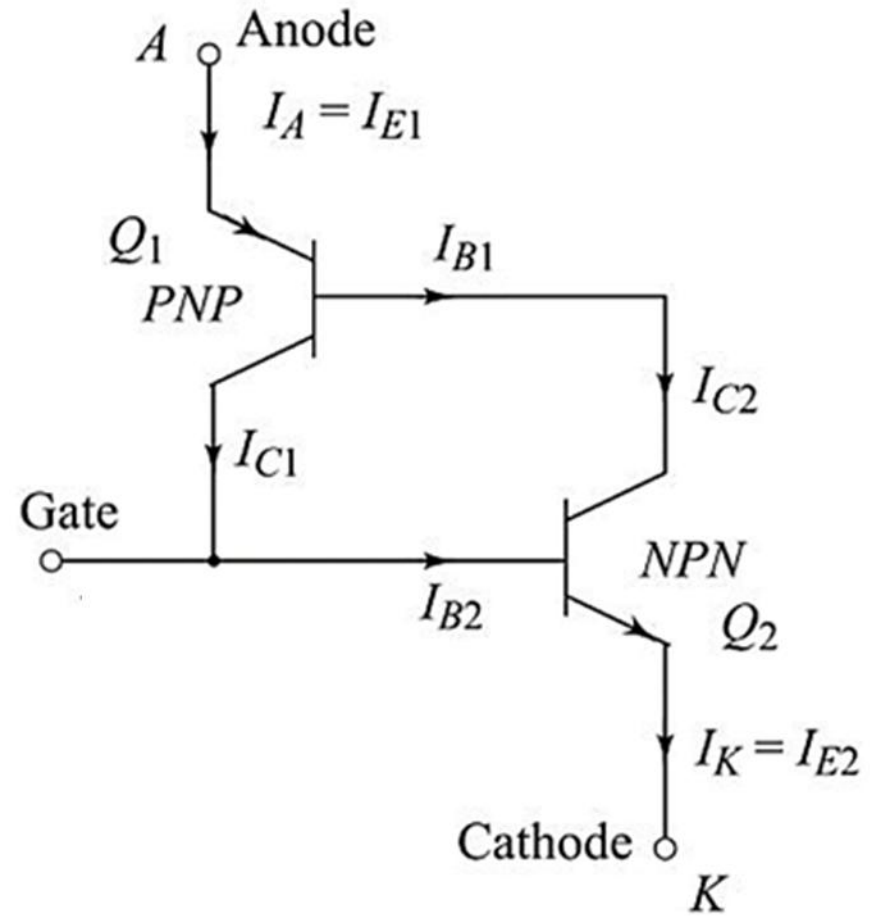
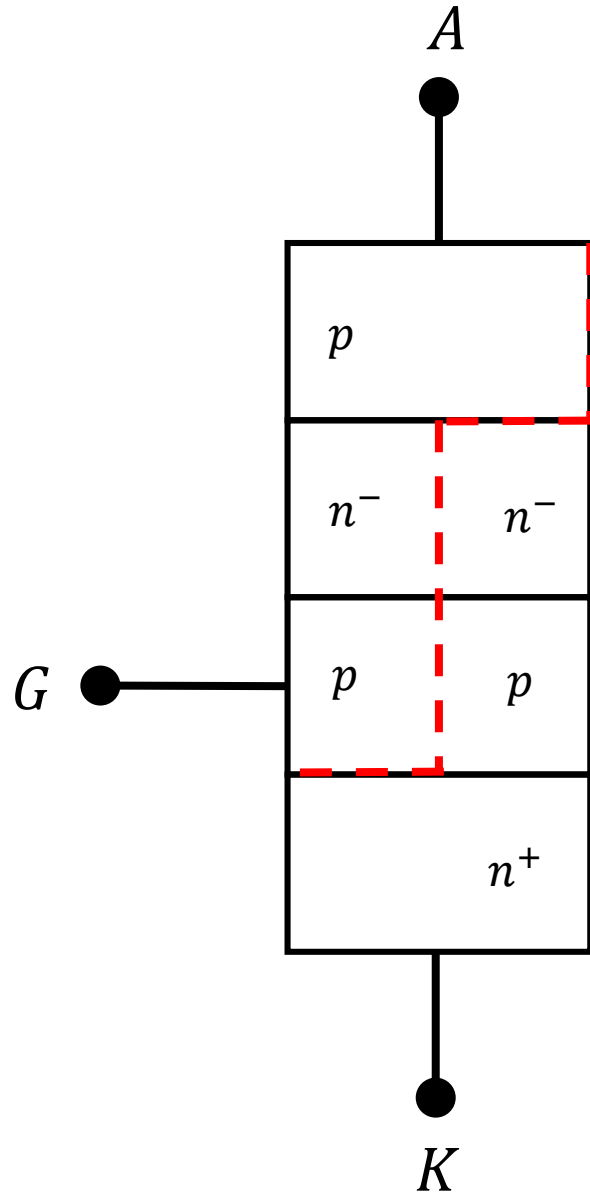
Turn off Time

- ❖ The turn-off time t_q of a thyristor is defined as the time between the instant anode current becomes zero and the instant SCR regains forward blocking capability.
- ❖ During time t_q , all the excess carriers from the four layers of SCR must be removed.
- ❖ The turn-off time is divided into two intervals; reverse recovery time t_{rr} and the gate recovery time t_{gr} ;

$$i.e. t_q = t_{rr} + t_{gr}$$
- ❖ The reverse recovery current removes excess carriers from the end junctions J1 and J3.
- ❖ At the end of reverse recovery period, the middle junction J2 still has trapped charges, therefore, the thyristor is not able to block the forward. These trapped charges must decay only by recombination.
- ❖ The time for the recombination of charges is called gate recovery time t_{gr} .



Two transistor analogy of SCR



Two transistor analogy of SCR

For transistor Q1

$$I_{C1} = \alpha_1 I_{E1} + I_{CBO1} \Rightarrow I_{C1} = \alpha_1 I_A + I_{CBO1}$$

For transistor Q2

$$I_{C2} = \alpha_2 I_{E2} + I_{CBO2} \Rightarrow I_{C2} = \alpha_2 I_K + I_{CBO2}$$

α_1 AND α_2 : CURRENT GAIN

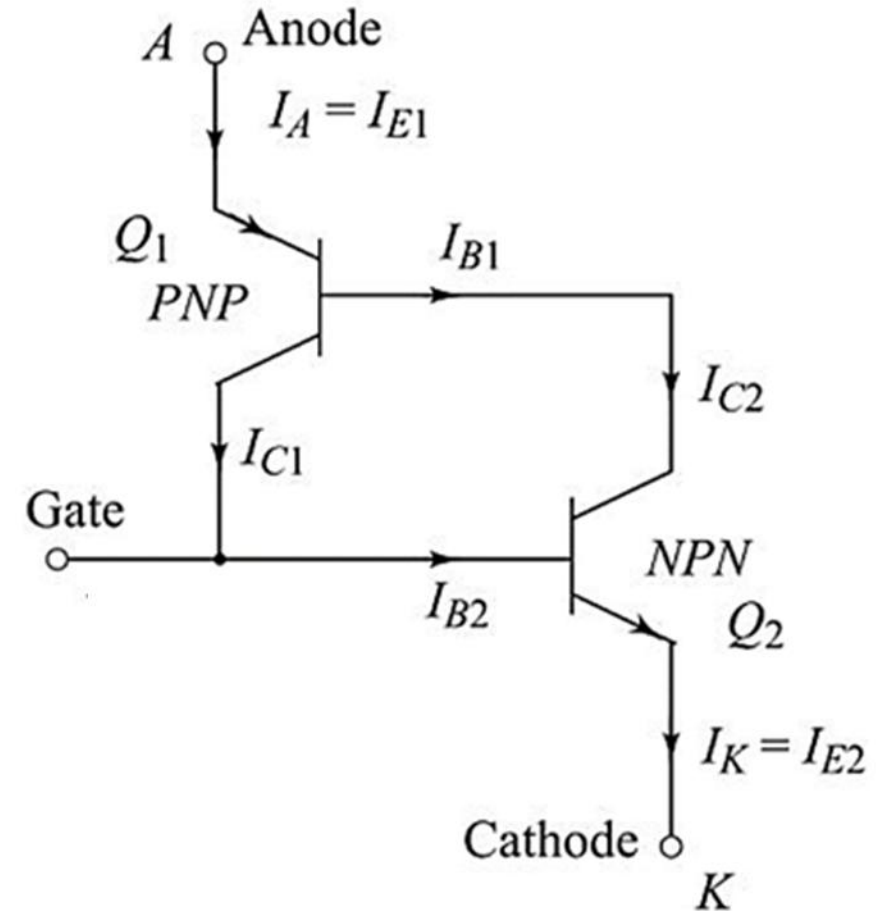
I_{CBO1} AND I_{CBO2} : LEAKAGE CURRENT

Also for Q1,

$$I_{E1} = I_{C1} + I_{B1}$$

$$I_A = I_{C1} + I_{C2} = \alpha_1 I_A + I_{CBO1} + \alpha_2 I_K + I_{CBO2}$$

$$I_A = \alpha_1 I_A + I_{CBO1} + \alpha_2 I_K + I_{CBO2} \rightarrow (1)$$



Two transistor analogy of SCR

$$I_A = \alpha_1 I_A + I_{CBO_1} + \alpha_2 I_K + I_{CBO_2} \rightarrow (1)$$

Similarly for Q2,

$$I_{E_2} = I_{C_2} + I_{B_2}$$

$$I_K = I_{C_2} + I_{C_1} + I_G$$

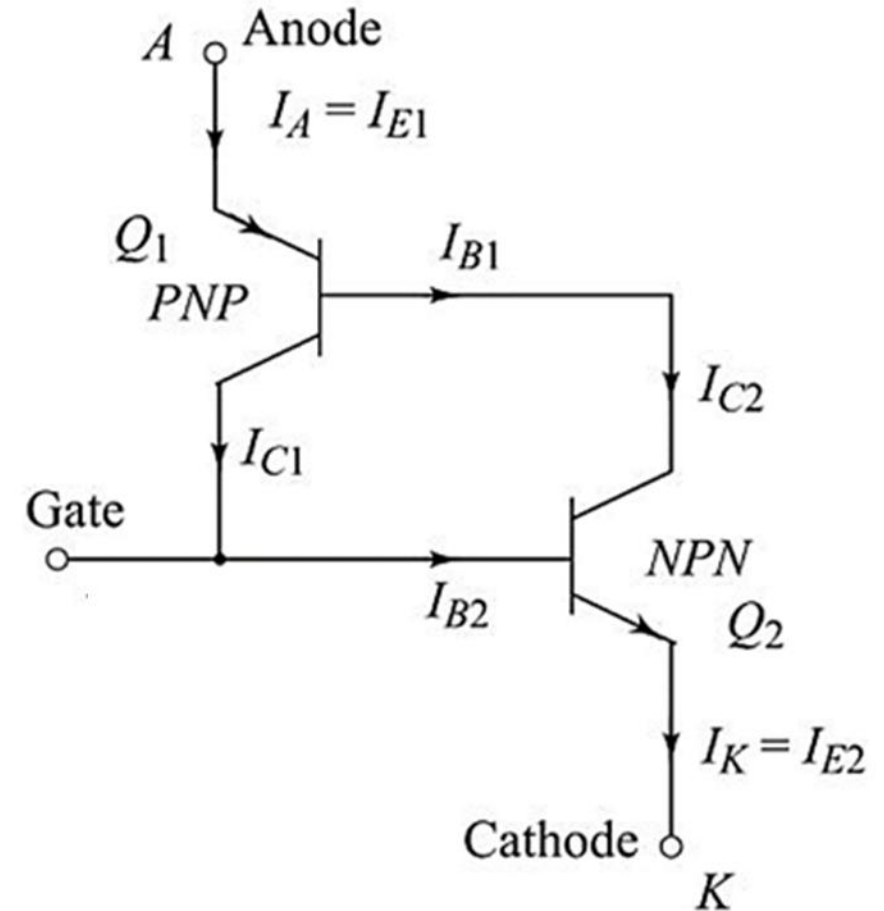
$$I_K = I_A + I_G \rightarrow (2)$$

Substituting (2) in equation (1)

$$I_A = \alpha_1 I_A + I_{CBO_1} + \alpha_2 (I_A + I_G) + I_{CBO_2}$$

$$I_A - \alpha_1 I_A - \alpha_2 I_A = \alpha_2 I_G + I_{CBO_1} + I_{CBO_2}$$

$$I_A = \frac{\alpha_2 I_G + I_{CBO_1} + I_{CBO_2}}{1 - (\alpha_1 + \alpha_2)}$$



Two transistor analogy of SCR

Current in the SCR is

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

Role of gate current

Case 1: when gate current $I_G = 0$, only leakage current flow in the SCR

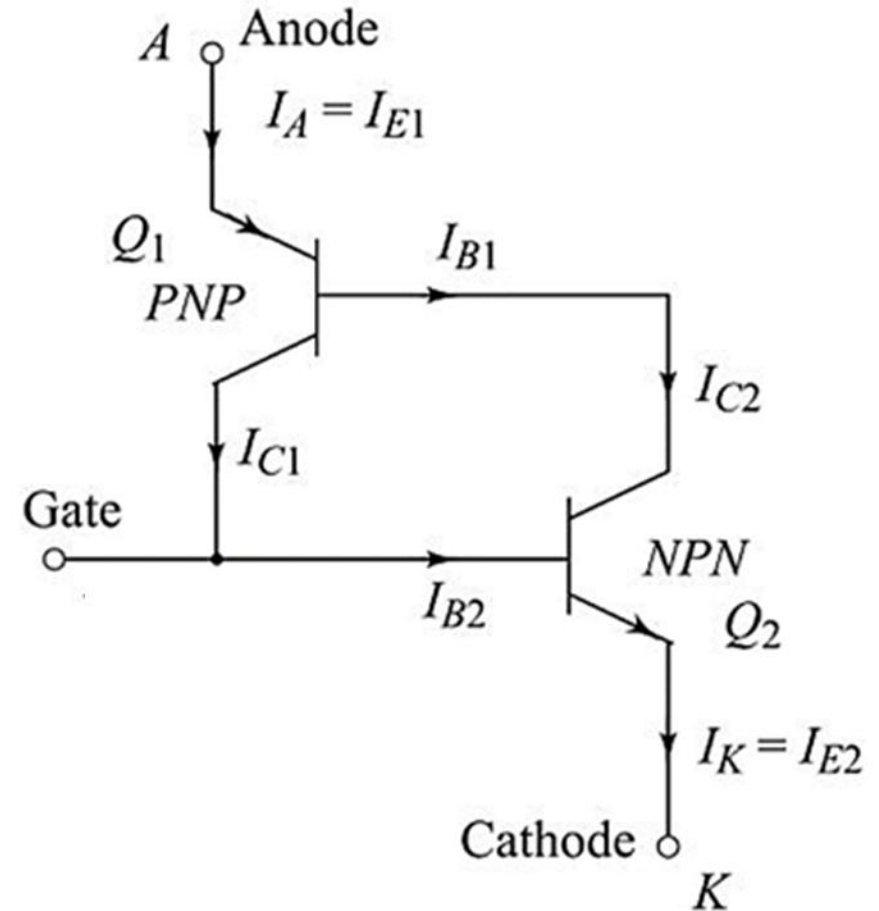
$$I_A = \frac{I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

In this case, as Q_2 is OFF, the value of $\alpha_2 \sim 0$

Similarly, as Q_1 is also OFF, the value of $\alpha_1 \sim 0$

$$I_A \sim I_{CBO1} + I_{CBO2}$$

So the SCR is OFF



Two transistor analogy of SCR

Current in the SCR is

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

Role of gate current

Case 2: when gate current is applied for a sufficient time,

$$I_{B2} = I_{C1} + I_G$$

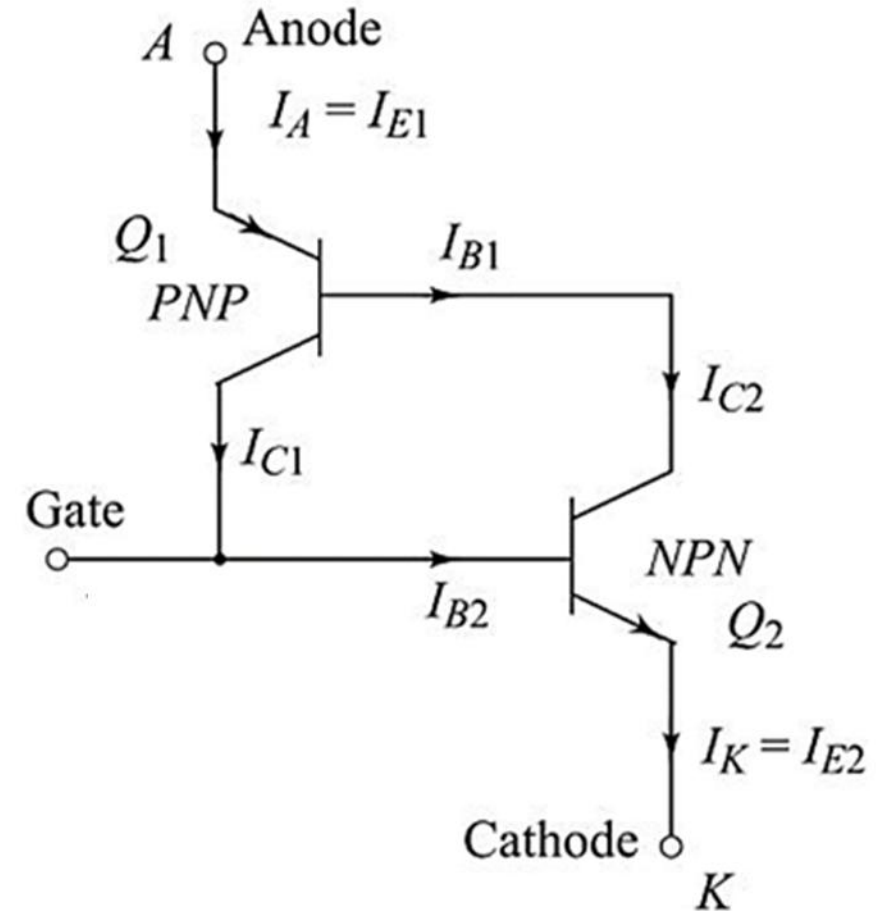
This base current will make Q2 ON, the value of $\alpha_2 > 0$

as Q2 is OFF, I_{C2} or I_{B1} will increase and the value of α_2 will no longer be zero and will increase. As I_{B1} will increase, this will turn on Q1

as Q1 is ON, I_{C1} will increase α_1 will no longer be zero and will increase. This further increase the value of I_{B2}

This regenerative process will continue and SCR will be ON. Also, due to this, $(\alpha_1 + \alpha_2)$ will tend to unity.

The current in the SCR will be only controlled by the external resistance in this case.



Two transistor analogy of SCR

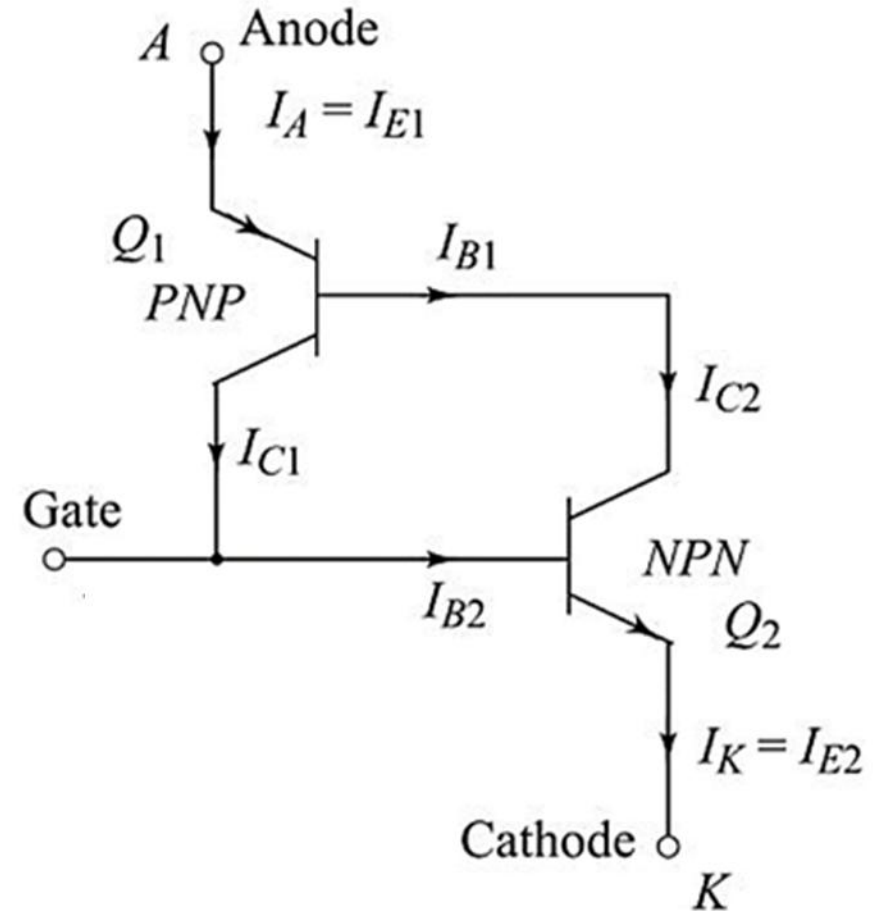
Current in the SCR is

$$I_A = \frac{\alpha_2 I_G + I_{CBO1} + I_{CBO2}}{1 - (\alpha_1 + \alpha_2)}$$

This regenerative process will continue and SCR will be ON. Also, due to this, $(\alpha_1 + \alpha_2)$ will tend to unity.

The current in the SCR will be only controlled by the external resistance in this case.

After applying gate current as both Q1 and Q2 will be ON and the collector current of one transistor will drive the base of the other transistor, the **SCR will not TURN OFF even after removal of GATE CURRENT.**



Triggering Methods of SCR

1. Forward Voltage Triggering (Breakover triggering)

Principle

- When the SCR is forward biased (anode positive w.r.t cathode), junctions J1 and J3 are forward biased, and J2 is reverse biased.
- If the anode–cathode voltage is increased beyond a certain value called the breakover voltage (V_{BO}):
 - The reverse-biased junction J2 undergoes avalanche breakdown.
 - This generates a large number of carriers.
 - Regenerative action starts between the internal PNP and NPN transistors.
 - SCR turns ON.

Features

- No gate signal required.
- Not preferred in practice because:
 - Requires high voltage.
 - May damage the device.
 - No control over the turn-on instant

2. Gate Triggering (Most common method)

Principle

- A positive voltage is applied to the gate with respect to the cathode.
- This injects carriers into the P-base region.
- It increases the current in the equivalent NPN transistor.
- Regenerative feedback between the two transistors causes the SCR to turn ON.

Features

- Most widely used method.
- Allows controlled turn-on at desired time.
- Requires relatively small gate current.

Types of gate triggering circuits

- 1. DC gate triggering**
 - Constant DC gate signal.
- 2. AC gate triggering**
 - Gate driven using AC source through phase control.
- 3. Pulse gate triggering**
 - Short pulses applied to gate.
 - Most efficient and commonly used in power electronics.

3. dV/dt Triggering

Principle

- When the anode–cathode voltage rises rapidly, The reverse-biased junction J2 behaves like a capacitor.
- Displacement current flows:

$$i = C \frac{dV}{dt}$$

- If this current becomes large enough, it acts like gate current.
- SCR turns ON unintentionally.

Features

- Unwanted triggering method.
- Occurs during fast voltage transients.

Prevention

- Use snubber circuits (RC network) across the SCR.

4. Thermal Triggering

Principle

- As temperature increases:
 - Leakage current across J2 increases.
 - This leakage current can act as gate current.
- If temperature rises enough:
 - SCR turns ON without gate signal.

Features

- Undesirable method.
- Occurs due to excessive heating.

5. Light Triggering (LASCR)

Principle

- In **Light Activated SCR (LASCR)**:
 - Light photons strike the silicon.
 - Electron–hole pairs are generated.
 - These carriers initiate regenerative action.
 - SCR turns ON.

Features

- High-voltage isolation.
- HVDC systems.
- Optical triggering systems.

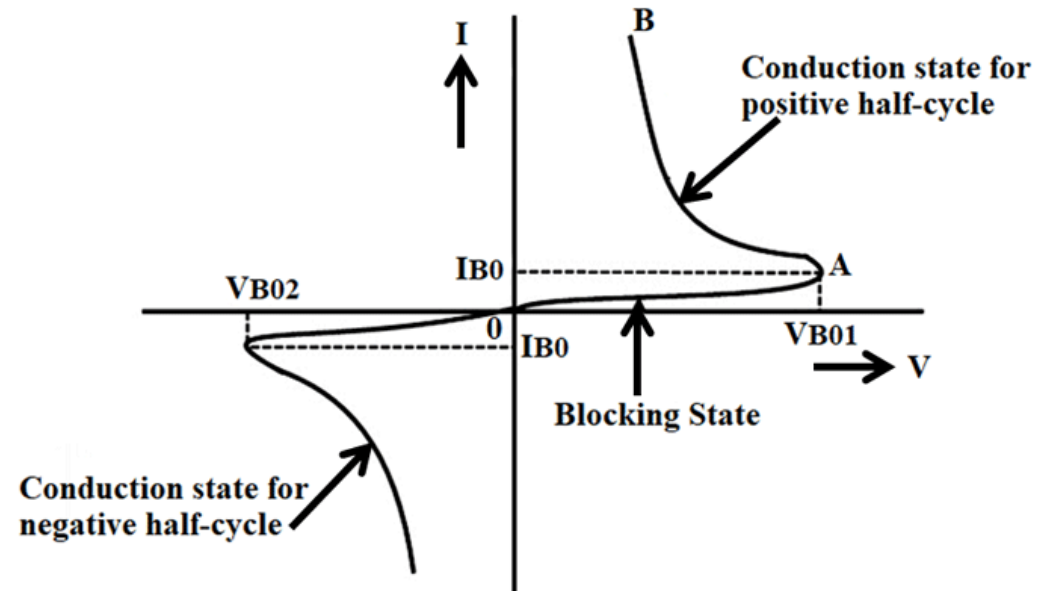
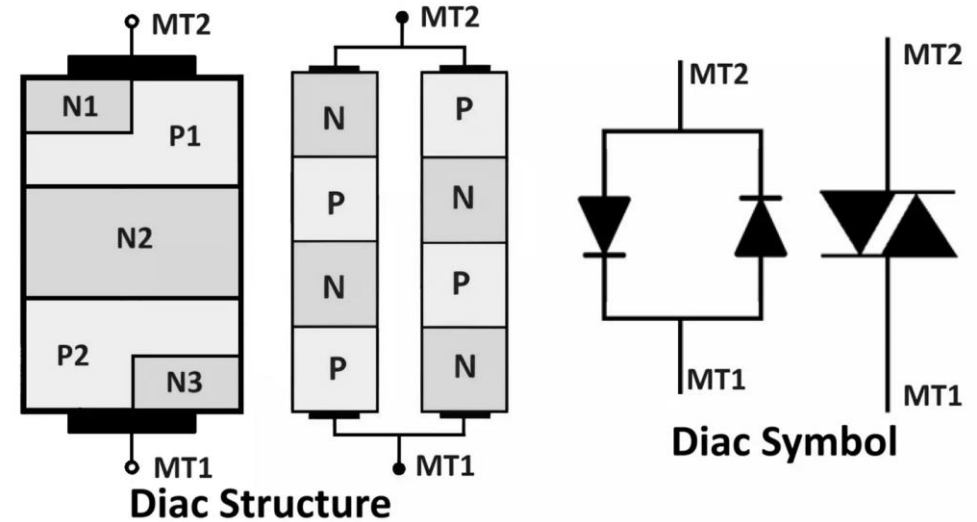
DIAC

- A DIAC is a special type of thyristor that is bidirectional and does not have a gate.
- Designed to be symmetrical, so it behaves the same in both directions of voltage.
- Bidirectional conduction:

For positive voltage at MT2 w.r.t MT1:
DIAC remains OFF until $+V_{BO}$ is reached.

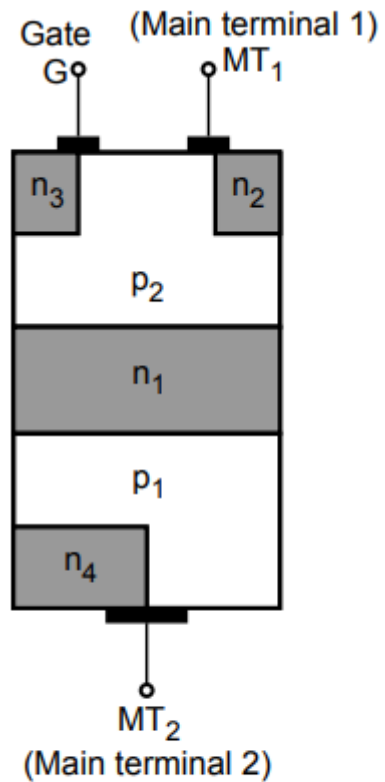
For negative voltage at MT2 w.r.t MT1:
DIAC remains OFF until $-V_{BO}$ is reached.

- Once conduction starts, DIAC has low voltage drop, and current flows until it falls below the holding current.

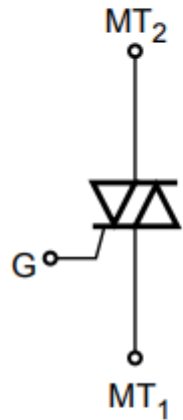


TRIAC

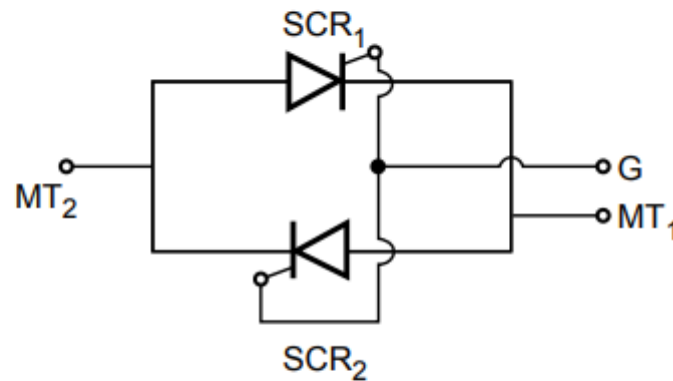
- Four-layer, three-junction PNPN structure, but effectively two SCRs connected in anti-parallel sharing a common gate.
- Designed to conduct in both directions, unlike an SCR which conducts in only one direction.
- Bidirectional conduction: Can conduct current from MT1 → MT2 or MT2 → MT1 when triggered.
- A small gate current can trigger conduction regardless of polarity of applied voltage.



(a) Construction



(b) Symbol



(c) Equivalent circuit

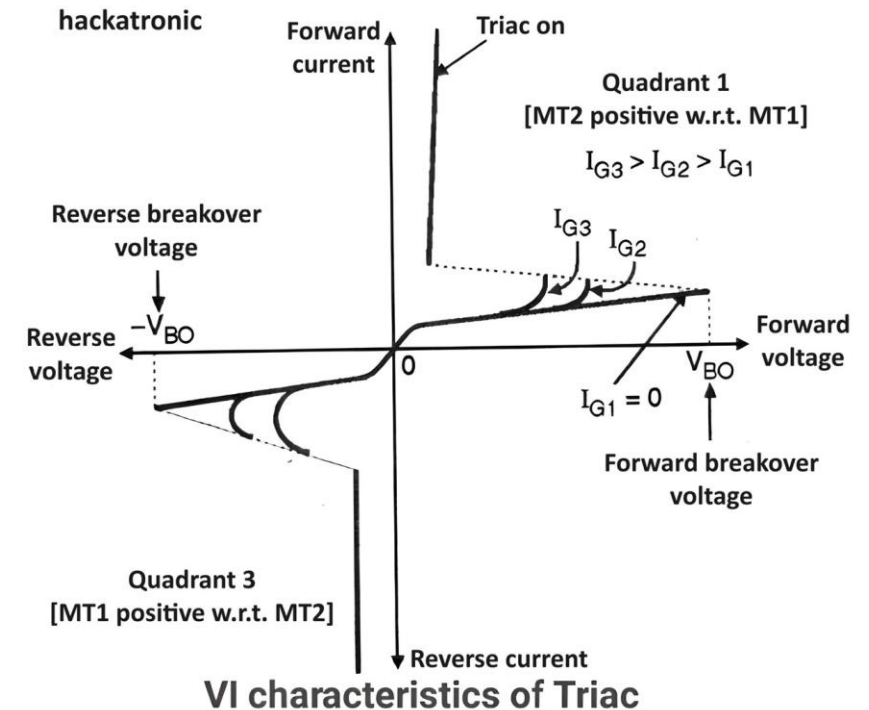
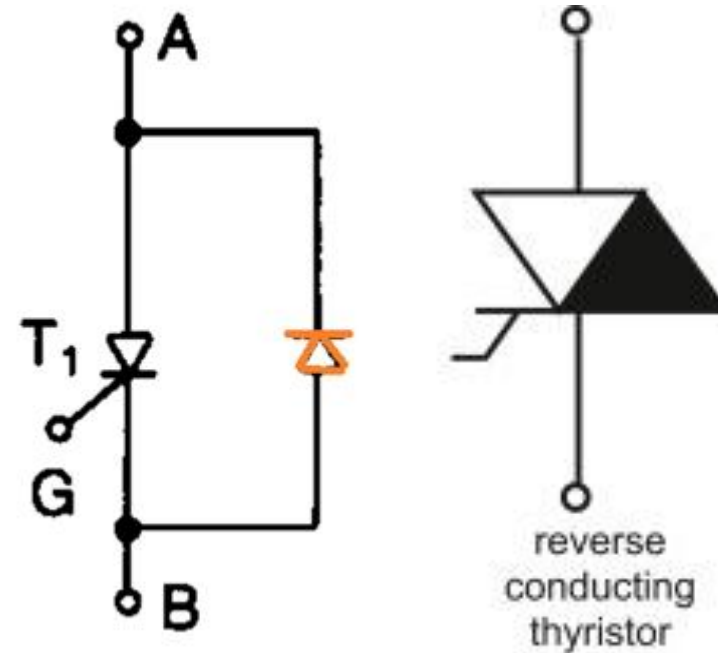


Fig. 5.5.1 Triac

ASCR/RCT



- Asymmetric design:
- Forward breakover voltage is low (turns ON easily).
- Reverse voltage withstand capability is high (blocks reverse voltage).



A Reverse Conducting Thyristor (RCT) is a combination of:
A standard SCR (for forward conduction)
A reverse diode (for reverse conduction)
These are integrated in a single package.

Purpose: To allow forward SCR conduction and safe reverse voltage handling in AC circuits.

LASCR

LASCR (Light Activated SCR)

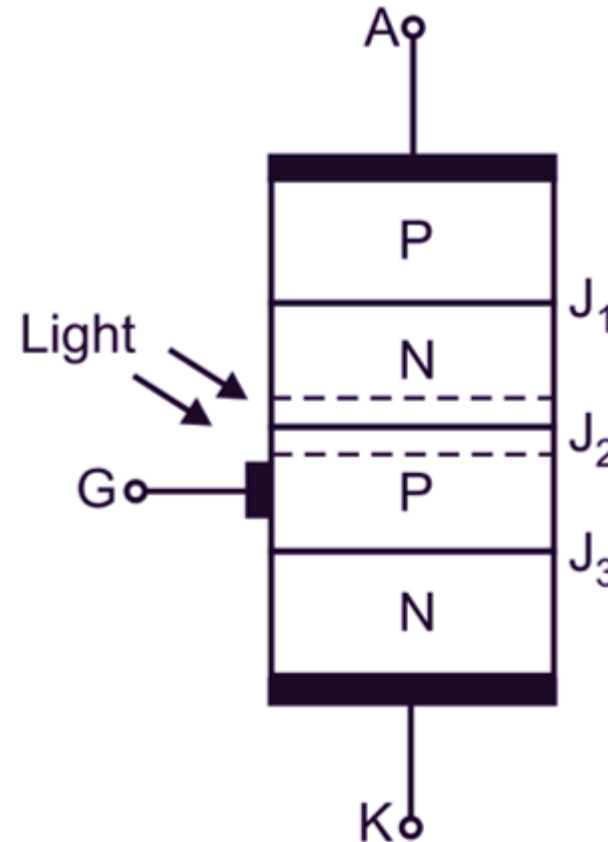
Structure: Same as SCR but with a light-sensitive gate region.

Triggering: Activated by light instead of electrical gate signal.

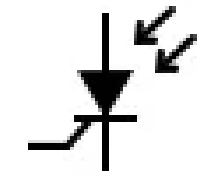
Advantages:

Electrical isolation between trigger and main circuit.

Applications: HVDC systems, isolated control in power electronics.

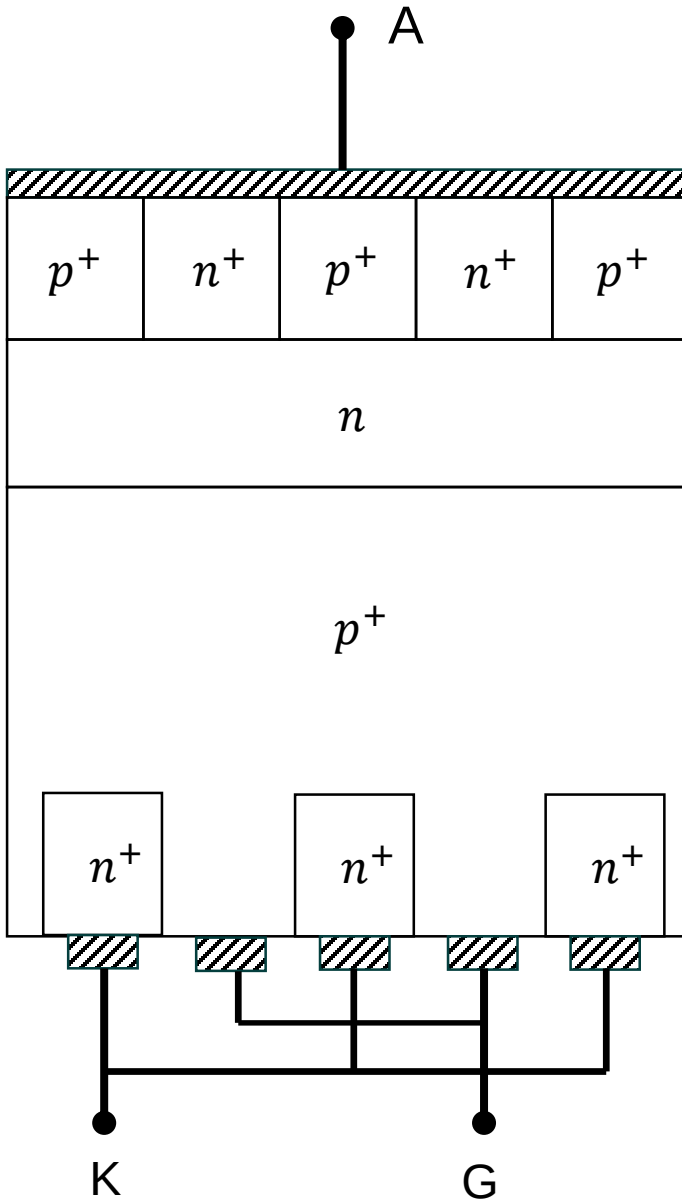


Light Activated SCR



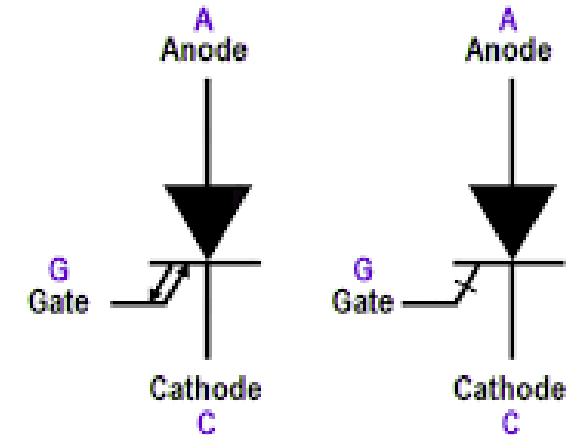
LASCR

Gate turn off thyristor (GTO)



- It is called GTO because it can be turned off by applying negative gate pulse
- To turn off GTO, we need gate current of the same order as anode current

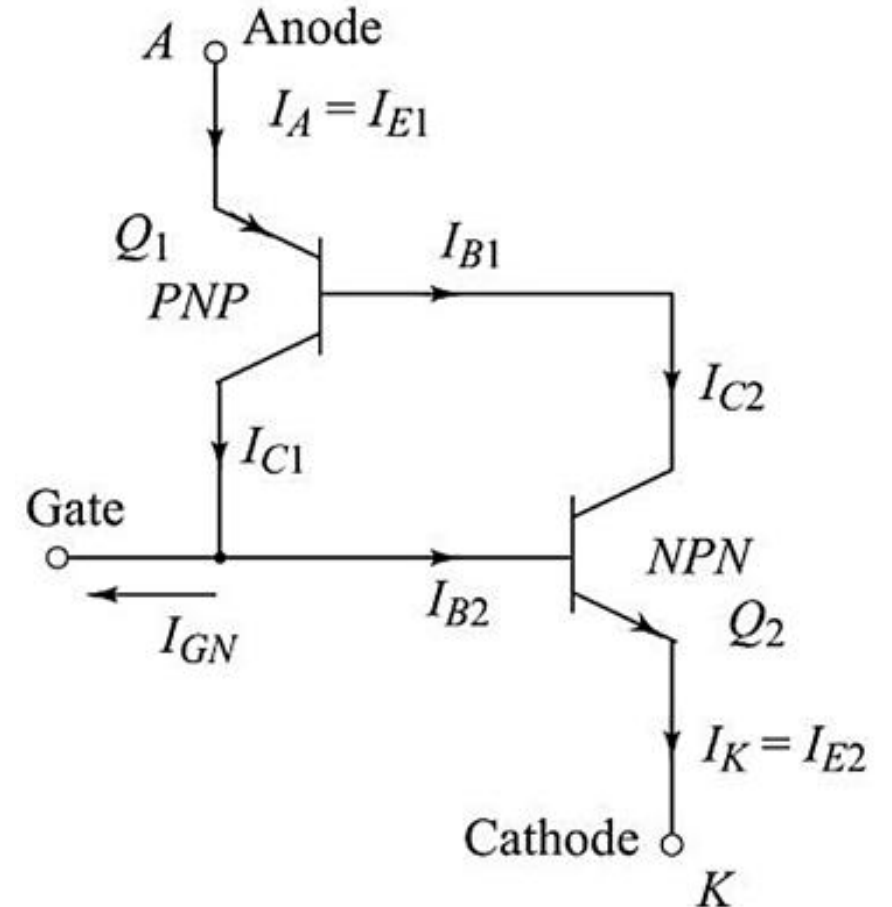
- n^+ fingers in anode
- Highly doped gate region
- Multi-cathode structure



Gate Turn-Off (GTO) Thyristor
Symbol

GTO Turn OFF mechanism

- A large negative gate current I_{GN} of the same order of anode current is applied.
- I_{C1} gets diverted towards gate terminal
- Base drive gets removed $\rightarrow$ gradually Q2 gets turned OFF
- Q2 OFF $\rightarrow$
- I_{C2} decreases $\rightarrow$
- I_{B1} decreases $\rightarrow$
- I_A decreases and gradually becomes Zero.



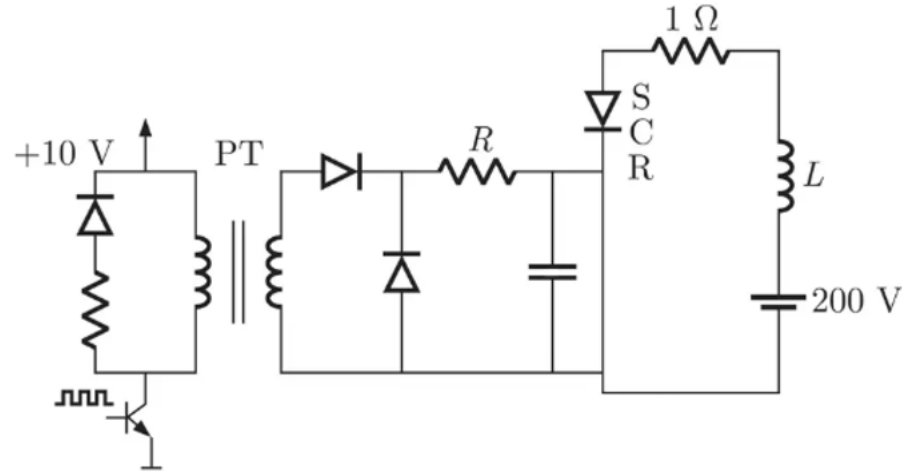
Feature	GTO (Gate Turn-Off Thyristor)	SCR (Silicon-Controlled Rectifier)
Turn-Off Mechanism	Can be turned off by applying a negative gate current	Cannot be turned off by gate signal, requires external circuit (e.g., commutation circuit)
Turn-On Mechanism	Requires a positive gate pulse	Requires a positive gate pulse
Control	Fully controllable (turn-on and turn-off)	Semi-controllable (only turn-on control)
Commutation	Self-commutated	Requires external commutation circuit
Efficiency	Higher efficiency due to controlled turn-off	Lower efficiency due to external commutation losses
Power Handling	Lower than SCR for the same rating	Higher power handling capability
Current and Voltage Rating	Typically, lower than SCR	Higher current and voltage ratings
Applications	Used in high-power inverters, motor drives, and traction systems	Used in rectifiers, controlled rectifiers, and power control applications
Losses	Higher conduction and switching losses	Lower conduction losses

Q1. For a power diode, the reverse recovery time is $3.9 \mu\text{s}$ and the rate of diode current decay is $50\text{A}/\mu\text{s}$. For a softness factor of 0.3 calculate the peak reverse recovery current and the stored charge.

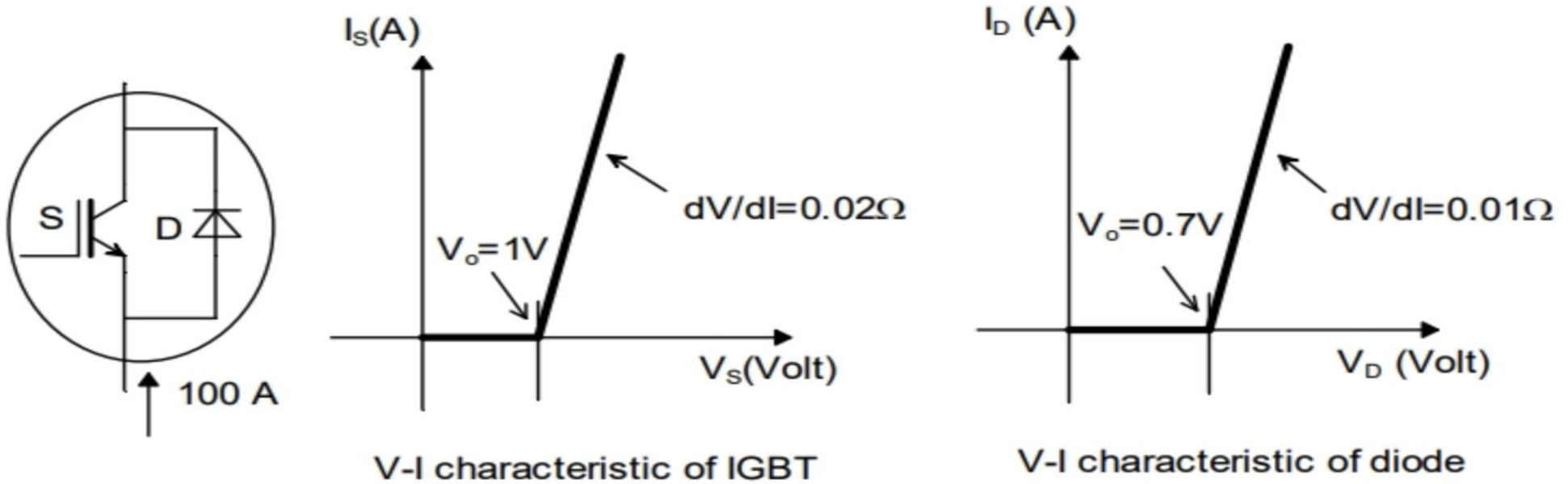
Q2. A 1:1 pulse transformer (PT) is used to trigger an SCR in the figure given below. The SCR is rated at 1.5 kV, 250 A with $I_L = 250 \text{ mA}$, $I_{G\text{max}} = 150 \text{ mA}$, $I_{G\text{min}} = 100 \text{ mA}$. The SCR is connected to an inductive load where $L = 150 \text{ mH}$ in series with a small resistance and the supply voltage is 200V dc. The forward voltage drop across all transistors/diodes and gate-cathode junction during ON state are 1.0 V.

(a) Calculate the value of R

(b) Calculate the maximum approximate voltage-second rating of the PT suitable for triggering the SCR.
(voltage-second rating is the maximum product of the voltage and the width of the pulse that may be applied)



Q.3 A steady dc current of 100 A is flowing through a power module (S, D) as shown in Figure (a). The V-I characteristics of the IGBT (S) and the diode (D) are shown in Figures (b) and (c), respectively. Calculate the conduction power loss in the power module (S, D).



Q1. For a power diode, the reverse recovery time is $3.9 \mu\text{s}$ and the rate of diode current decay is $50\text{A}/\mu\text{s}$. For a softness factor of 0.3 calculate the peak reverse recovery current and the stored charge.

$$I_{RR} = \frac{di}{dt} * t_a$$

$$t_a = t_{rr} - t_b$$

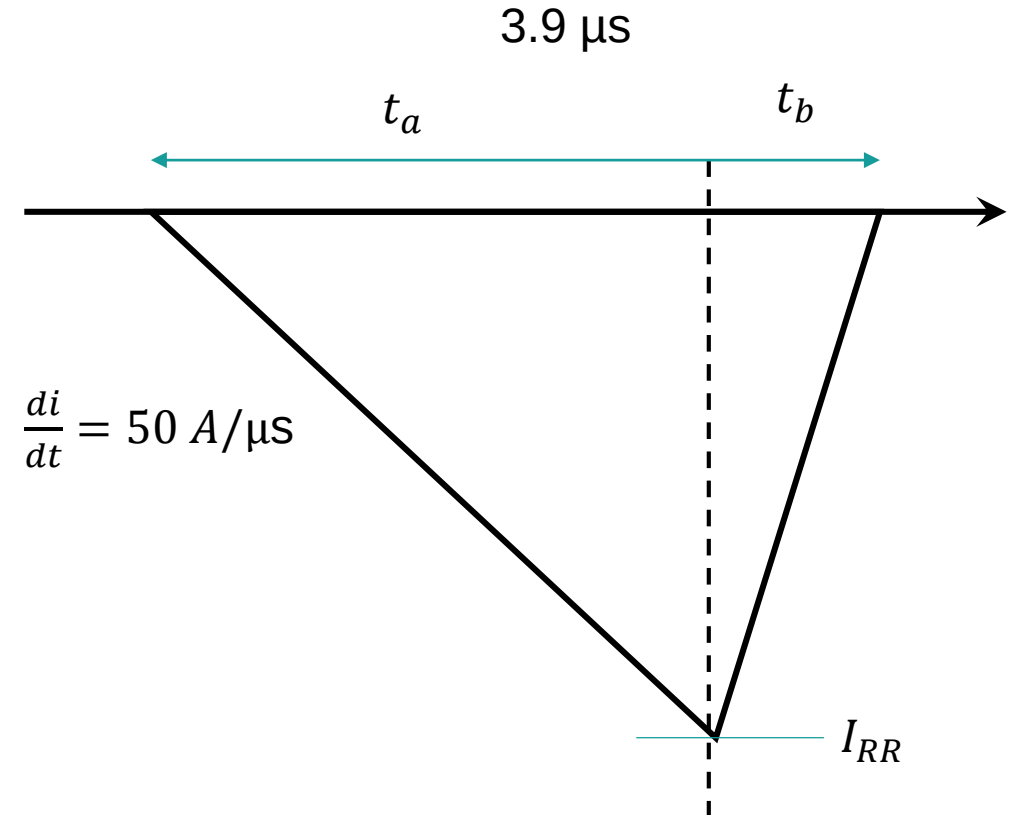
$$\frac{t_b}{t_a} = 0.3 \Rightarrow t_b = 0.3 t_a$$

$$t_a = t_{rr} - 0.3t_a$$

$$t_a = \frac{t_{rr}}{1.3} = \frac{3.9}{1.3} = 3\mu\text{s}$$

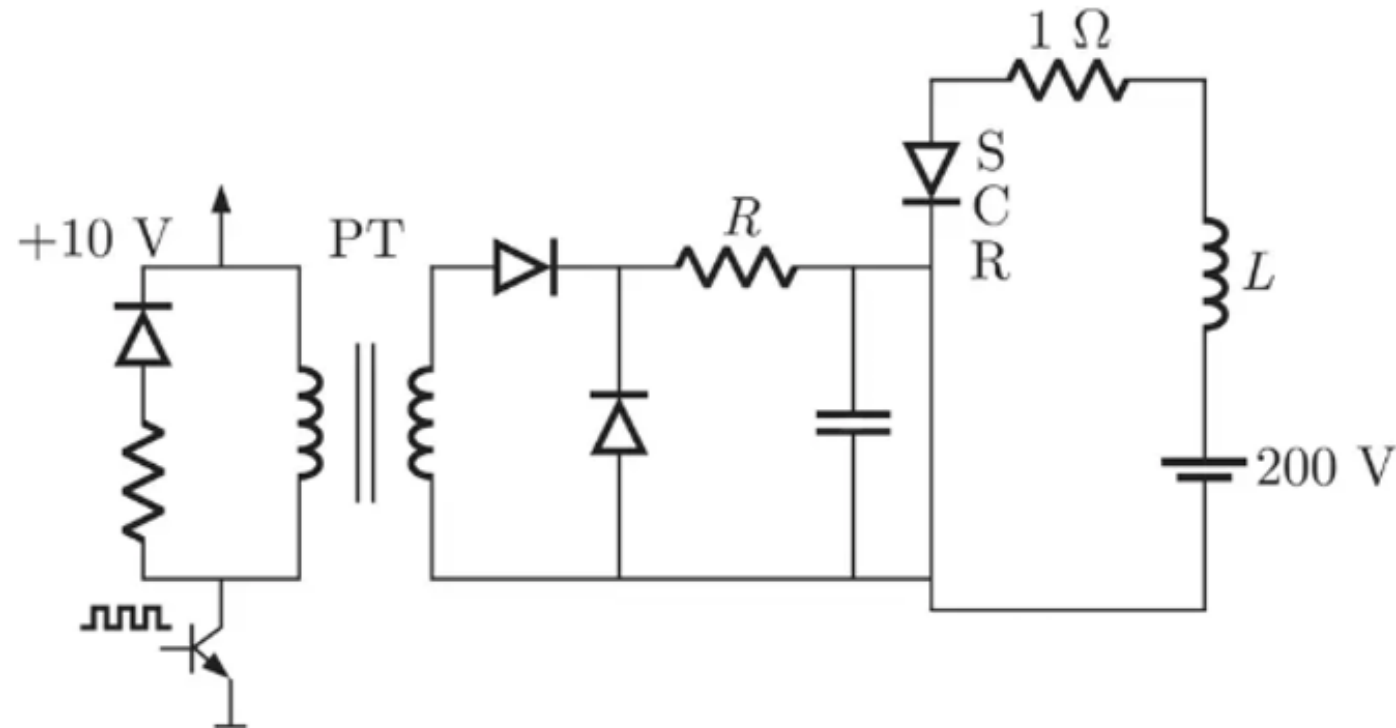
$$I_{RR} = 50 * 3 = 150 \text{ A}$$

$$Q_{RR} = \frac{1}{2} * I_{RR} * t_{rr} = 0.5 * 150 * 3.9 \mu\text{C} = 292.5 \mu\text{C}$$



Q2. A 1:1 pulse transistor (PT) is used to trigger and SCR in the figure given below. The SCR is rated at 1.5 kV, 250 A with $I_L = 250 \text{ mA}$, $I_{Gmax} = 150 \text{ mA}$, $I_{Gmin} = 100 \text{ mA}$. The SCR is connected to an inductive load where $L = 150 \text{ mH}$ in series with a small resistance and the supply voltage is 200V dc. The forward voltage drop across all transistors/diodes and gate-cathode junction during ON state are 1.0 V.

- Calculate the value of R
- Calculate the maximum approximate voltage-second rating of the PT suitable for triggering the SCR.
(voltage-second rating is the maximum product of the voltage and the width of the pulse that may be applied)



$$-10 + V_1 + V_{CE} = 0$$

$$V_{CE} = 1 \text{ V}$$

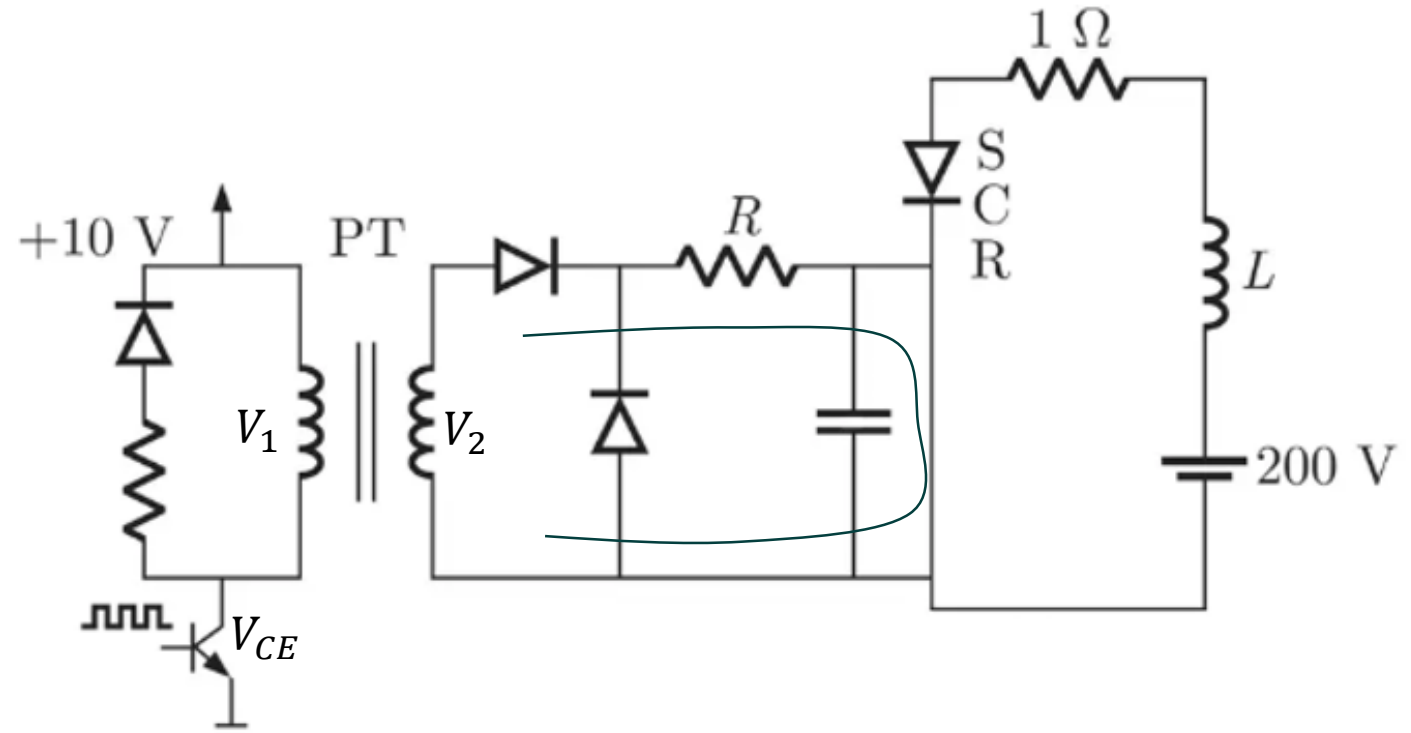
$$V_1 = 10 - 1 = 9 \text{ V}$$

$$V_2 = V_1 = 9 \text{ V}$$

$$-V_2 + V_d + V_r + V_{AK} = 0$$

$$V_r = V_2 - V_d - V_{AK} = 9 - 1 - 1 = 7 \text{ V}$$

$$V_r = I_G R \Rightarrow R = \frac{V_r}{I_G} = 46.67 \text{ } \Omega$$

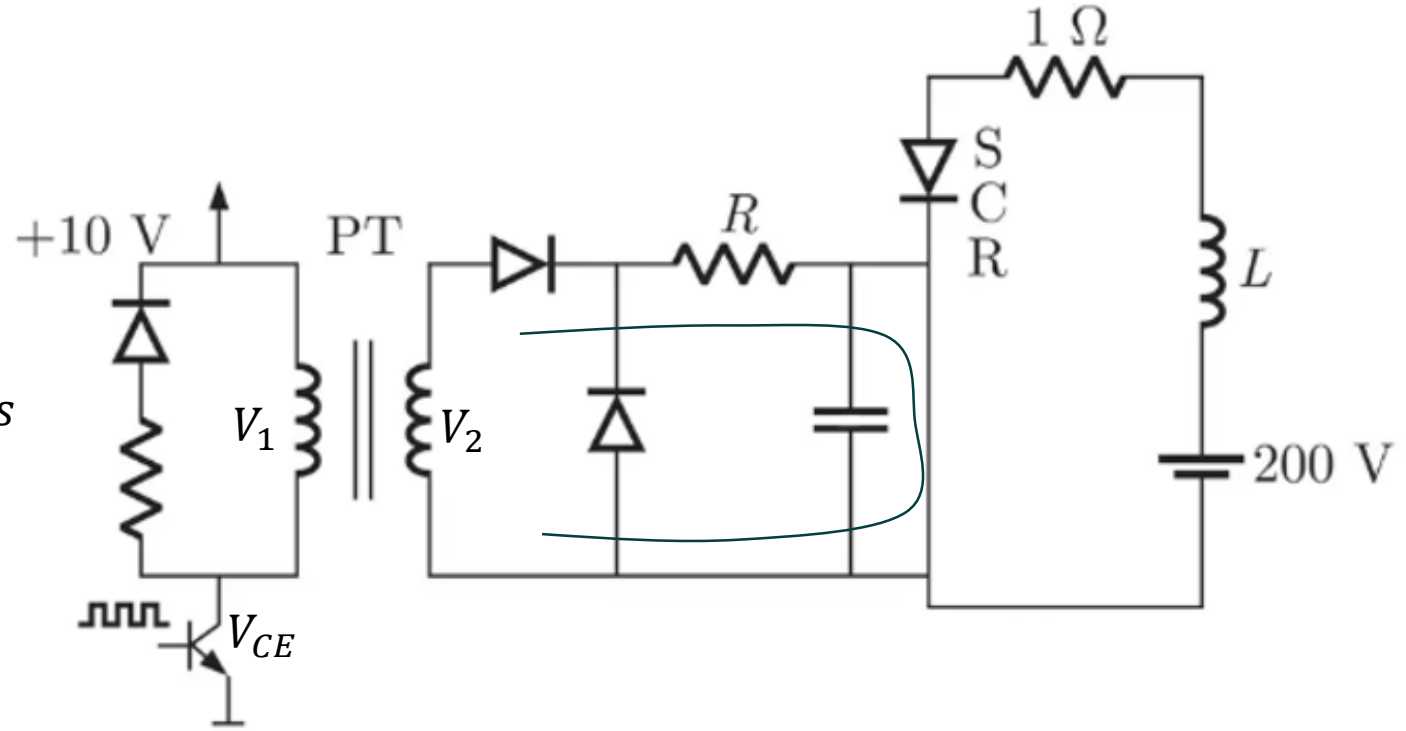


$$i_L = \frac{V}{R} (1 - e^{-\frac{t}{\frac{L}{R}}})$$

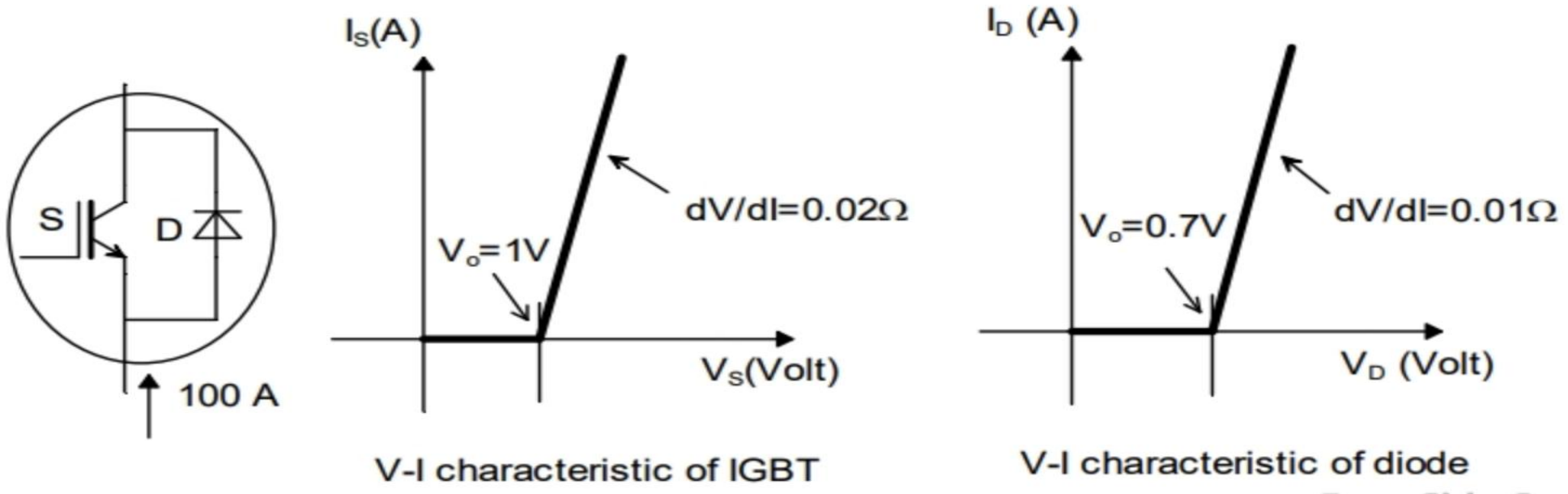
$$0.25 = 200(1 - e^{-\frac{t}{0.15}})$$

$$t = 188.56 \mu s$$

$$V - s \text{ rating} = V_m * t = 10 * 188.56 \approx 2000 \mu V - s$$



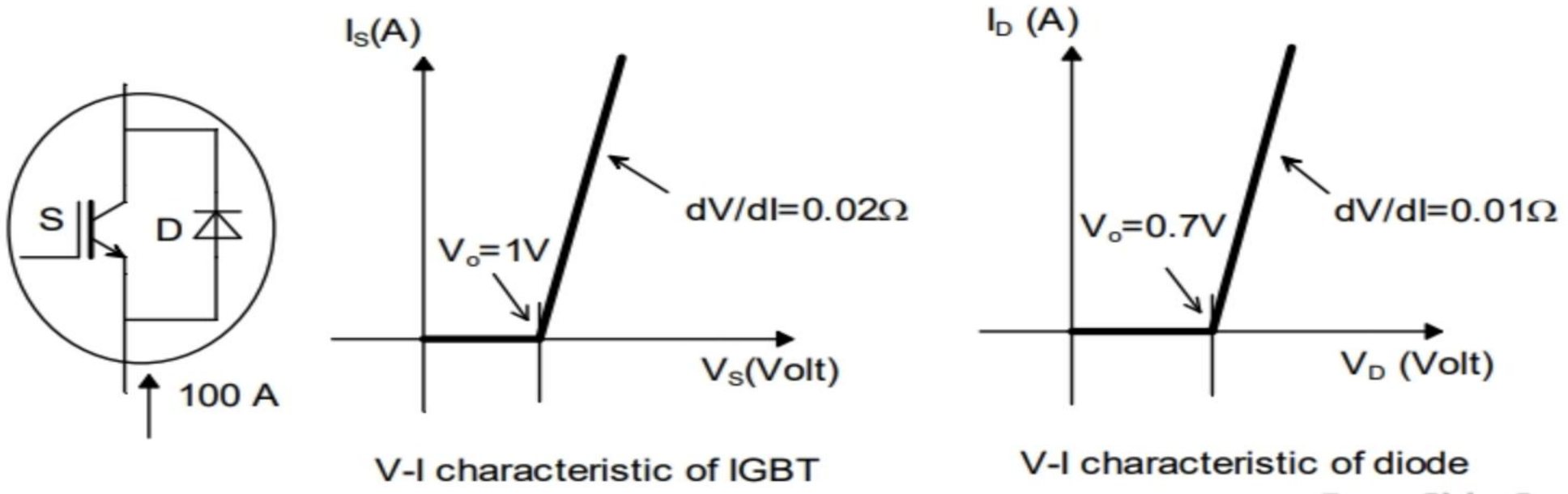
Q.3 A steady dc current of 100 A is flowing through a power module (S, D) as shown in Figure (a). The V-I characteristics of the IGBT (S) and the diode (D) are shown in Figures (b) and (c), respectively. Calculate the conduction power loss in the power module (S, D).



$$\text{Conduction loss} = v_{on} * i_{on}$$

In the above module current direction indicates that the current is flowing only through the diode.

Q.3 A steady dc current of 100 A is flowing through a power module (S, D) as shown in Figure (a). The V-I characteristics of the IGBT (S) and the diode (D) are shown in Figures (b) and (c), respectively. Calculate the conduction power loss in the power module (S, D).



$$\text{Conduction loss} = v_{on} * i_{on}$$

Also diode is modelled as a combination of on-state voltage drop in series with resistance.

$$\text{Conduction loss} = v_d * i_{on} + v_R * i_{on} = 0.7 * 100 + 100^2 * 0.01 = 70 + 100 = 170 \text{ W}$$